Irrational Numbers and Their Location on the Number Line

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Rational numbers can be written as a ratio of two integers, such as 3, -5, and $\frac{7}{8}$.

Irrational Numbers

Irrational numbers cannot be expressed as a ratio of two integers.
Their decimal forms are **non-terminating** and **non-repeating**.

Square Roots

A square root such as $\sqrt{n}$ is **irrational** when n is not a perfect square.

$$\sqrt{2}, \quad \sqrt{7}, \quad \sqrt{10}$$

are irrational numbers.

Cube Roots

A cube root such as $\sqrt[3]{n}$ is **irrational** when n is not a perfect cube.

$$\sqrt[3]{5}$$

is an irrational number.

Locating Irrational Numbers

To locate an irrational number on a **number line**, determine the two consecutive integers between which it lies.
Since

$$2^2 < 5 < 3^2,$$

we have

$$2 < \sqrt{5} < 3.$$

Decimal Approximation

Irrational numbers can be approximated by decimals to help locate them more accurately on a number line.

$$\sqrt{5} \approx 2.236$$

Therefore, $\sqrt{5}$ lies between 2 and 3.

Examples

Example 1

Determine whether $\sqrt{12}$ is rational or irrational.

Since 12 is not a perfect square,

$$\sqrt{12} \text{ is irrational.}$$

Example 2

Determine whether $\sqrt{81}$ is rational or irrational.
Since

$$81 = 9^2,$$

$$\sqrt{81} = 9$$

so it is **rational**.

Example 3

Between which two consecutive integers does $\sqrt{15}$ lie?

Since

$$3^2 < 15 < 4^2,$$

we have

$$3 < \sqrt{15} < 4.$$

Practice Exercises

Practice 1

Determine whether $\sqrt{12}$ is rational or irrational.

Practice 2

Determine whether $\sqrt{81}$ is rational or irrational.

Practice 3

Determine whether $\sqrt[3]{18}$ is rational or irrational.

Practice 4

Determine whether $\sqrt[3]{125}$ is rational or irrational.

Practice 5

Between which two consecutive integers does $\sqrt{15}$ lie?

Practice 6

Between which two consecutive integers does $\sqrt{30}$ lie?

Practice 7

Between which two consecutive integers does $\sqrt[3]{20}$ lie?

Practice 8

Between which two consecutive integers does $\sqrt[3]{50}$ lie?

Practice 9

Arrange from least to greatest:

$$\sqrt{3}, \quad 2, \quad \sqrt{8}$$

Practice 10

Arrange from least to greatest:

$$\sqrt[3]{4}, \quad 1, \quad \sqrt[3]{10}$$

Activities

Activity 1

Determine whether $\sqrt{27}$ is rational or irrational.

Activity 2

Determine whether $\sqrt{144}$ is rational or irrational.

Activity 3

Determine whether $\sqrt[3]{17}$ is rational or irrational.

Activity 4

Determine whether $\sqrt[3]{216}$ is rational or irrational.

Activity 5

Between which two consecutive integers does $\sqrt{22}$ lie?

Activity 6

Between which two consecutive integers does $\sqrt{40}$ lie?

Activity 7

Between which two consecutive integers does $\sqrt[3]{15}$ lie?

Activity 8

Between which two consecutive integers does $\sqrt[3]{70}$ lie?

Activity 9

Arrange from least to greatest:

$$\sqrt{5}, \quad 3, \quad \sqrt{11}$$

Activity 10

Arrange from least to greatest:

$$\sqrt[3]{2}, \qquad \sqrt[3]{12}, \qquad 2$$

Thank you!